EXCELLENCE

CLASS 11 · 2026–27

Physics | Lecture - 02

Basic Maths and Calculus

Height, ASTC and radian

By — Excellence Programme

2 September 2026

Topics to be covered

01 5-minute recap of Lecture 01 — ratios, identities, range, and last-class questions

02 Homework from last class — prove $1 + \tan^2\theta = \sec^2\theta$ and $1 + \cot^2\theta = \operatorname{cosec}^2\theta$

03 Height and distance — elevation and depression (leftover from Lecture 01)

04 ASTC — which ratios are positive in each quadrant

05 Standard angles 0° to 180°, and 37° / 53°

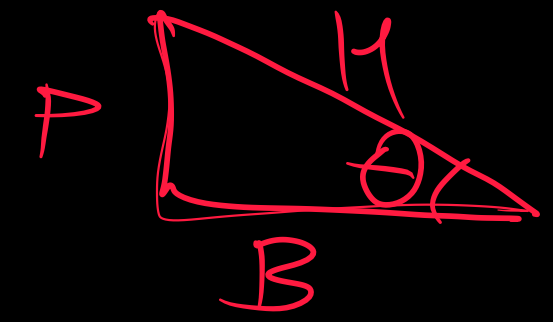
06 Radian measure, arc = Rθ, degree ↔ radian

Last class — Lecture 01

We stop here only to lock what you already wrote. Then today's leftover board.

P — opposite θ **B — adjacent to θ** **H — hypotenuse**

Sides are named from θ, not from how the figure sits.



sin θ = P/H **cos θ = B/H** **tan θ = P/B**

cosec θ = H/P **sec θ = H/B** **cot θ = B/P**

sin · cosec = 1 **cos · sec = 1** **tan · cot = 1**

$\sin\theta \times \frac{1}{\sin\theta} = 1$

$\cos\theta \cdot \frac{1}{\cos\theta} = 1$

$\tan\theta \cdot \frac{1}{\tan\theta} = 1$

The three identities — one Pythagoras

Start from $P^2 + B^2 = H^2$. Divide by H^2, by B^2, by P^2.

$$\sin^2\theta + \cos^2\theta = 1$$
$$1 + \tan^2\theta = \sec^2\theta$$
$$1 + \cot^2\theta = \operatorname{cosec}^2\theta$$

If you keep only one line: $\sin^2\theta + \cos^2\theta = 1$

$$\frac{\sin^2\theta + \cos^2\theta = 1}{\sin^2\theta} = \frac{\sin^2\theta}{\sin^2\theta} + \frac{\cos^2\theta}{\sin^2\theta} = \frac{1}{\sin^2\theta}$$

$$1 + \left(\frac{\cos\theta}{\sin\theta}\right)^2 = \operatorname{cosec}^2\theta$$

$$1 + \cot^2\theta = \operatorname{cosec}^2\theta$$

The three identities — one Pythagoras

Start from $P^2 + B^2 = H^2$. Divide by H^2, by B^2, by P^2.

1. $\sin^2\theta + \cos^2\theta = 1$ → ÷ $\cos^2\theta$
2. $1 + \tan^2\theta = \sec^2\theta$
3. $1 + \cot^2\theta = \operatorname{cosec}^2\theta$

$$\frac{\sin^2\theta + \cos^2\theta = 1}{\cos^2\theta}$$

$$\frac{\sin^2\theta}{\cos^2\theta} + \frac{\cos^2\theta}{\cos^2\theta} = \frac{1}{\cos^2\theta}$$

$$\tan^2\theta + 1 = \sec^2\theta$$

If you keep only one line: $\sin^2\theta + \cos^2\theta = 1$

Eqn → (2) & (3) are derived form of Eqn (1)

Range, and the questions we solved

Sine is P/H. H is longest, so sine cannot be 2.

$-1 \le \sin\theta \le 1$ **$-1 \le \cos\theta \le 1$** **$\tan 90°$ is not defined**

Questions from last class — you already have these in the notebook:

$\tan\theta = 2$ → $P = 2k$, $B = k$, $H = \sqrt{5}\,k$ $\sin = 2/\sqrt{5}$, $\cos = 1/\sqrt{5}$

$(\sin\theta + \cos\theta)/(\sin\theta - \cos\theta) = 11/5$ → $\tan\theta = 8/3$

Nested similar triangles → $y = 3/2$ $x = 8/3$

We stopped after x = 8/3. First: last-class homework. Then height and distance.

Total

Union & Intersection

Common

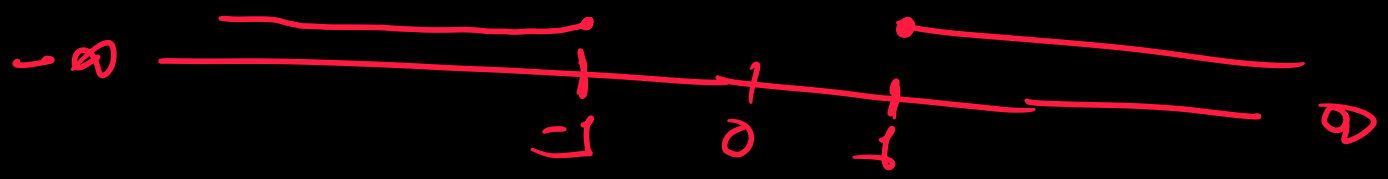
Range of $\operatorname{cosec}\theta$ → $(-\infty, -1] \cup [1, \infty)$

// to $-\sec\theta$

$\sin\theta$ $\cos\theta$ → same

$\operatorname{cosec}\theta$ & $\sec\theta$ → same

$\tan\theta$ & $\cot\theta$ → same



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"[" → repr. ⇒ value include in Range

"(" → Excluded

$\tan\theta \rightarrow [-\infty, \infty] \rightarrow$ wrong

$[-\infty, \infty)$

correct ← $(-\infty, \infty)$ → If Exclude both End values

Range, and the questions we solved

Sine is P/H. H is longest, so sine cannot be 2.

include

$-1 \le \sin\theta \le 1$ **$-1 \le \cos\theta \le 1$** **$\tan 90°$ is not defined**

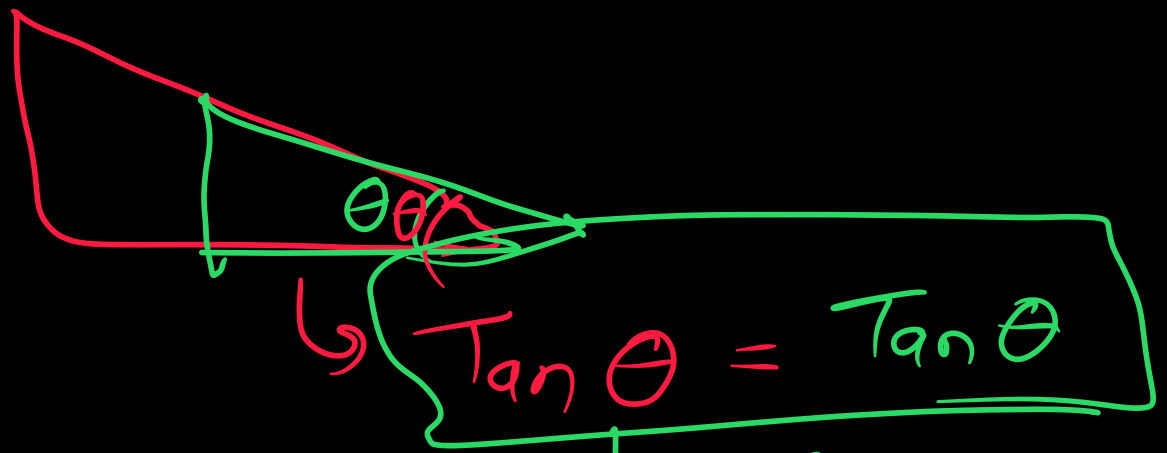
Questions from last class — you already have these in the notebook:

$\tan\theta = 2$ → **P = 2k, B = k, H = √5 k** **sin = 2/√5, cos = 1/√5**

$(\sin\theta + \cos\theta)/(\sin\theta - \cos\theta) = 11/5$ → **$\tan\theta = 8/3$**

Nested similar triangles → **y = 3/2** **x = 8/3**

We stopped after x = 8/3. First: last-class homework. Then height and distance.



↳ Same angle thier Edges are different

From last class — prove this identity

M.W

Starting from $P^2 + B^2 = H^2$, divide both sides by B^2 and hence prove that $1 + \tan^2\theta = \sec^2\theta$.

Copy the question. This is homework from last class. Do not wait for the proof on this board.

From last class — prove this identity

H.W

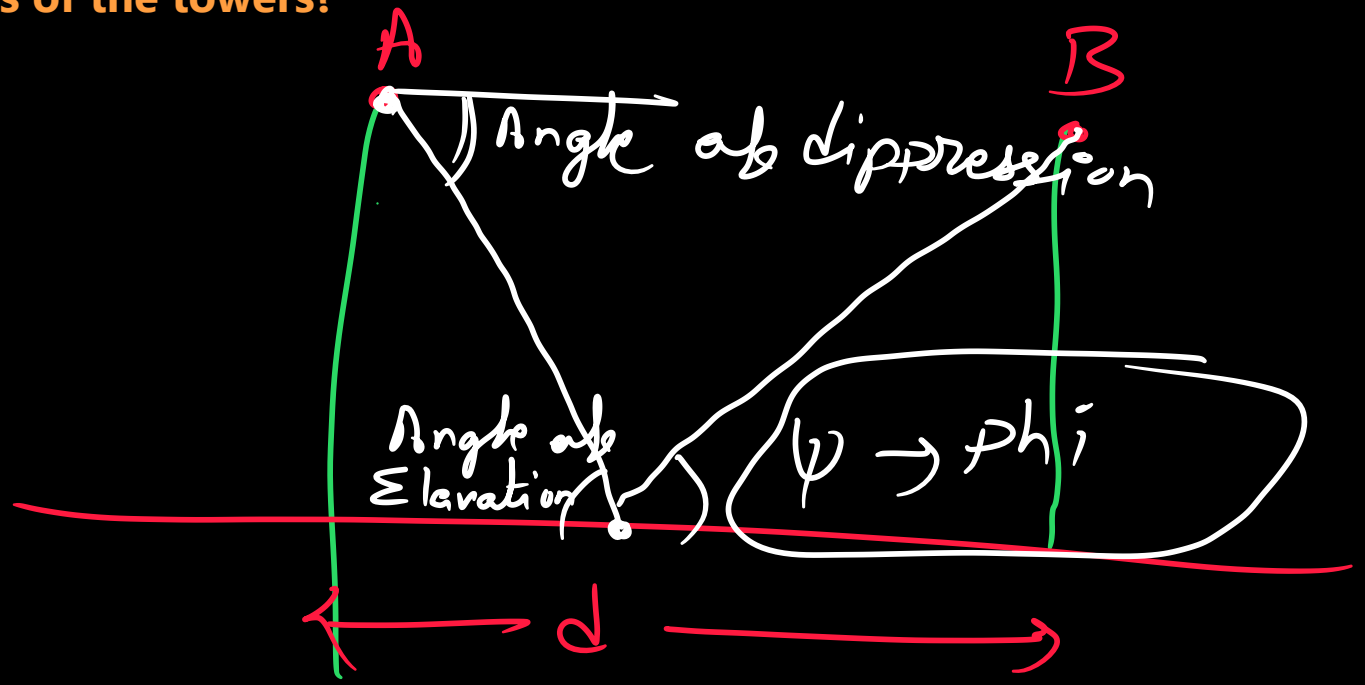
Starting from $P^2 + B^2 = H^2$, divide both sides by P^2 and hence prove that $1 + \cot^2\theta = \operatorname{cosec}^2\theta$.

Copy the question. This is homework from last class. Do not wait for the proof on this board.

Observer between two towers — elevation

From a point on level ground between two vertical towers that are d metres apart, the angles of elevation to the tops of the nearer and farther towers are θ and φ respectively. The observer is x metres from the nearer tower. What is the difference in heights of the towers?

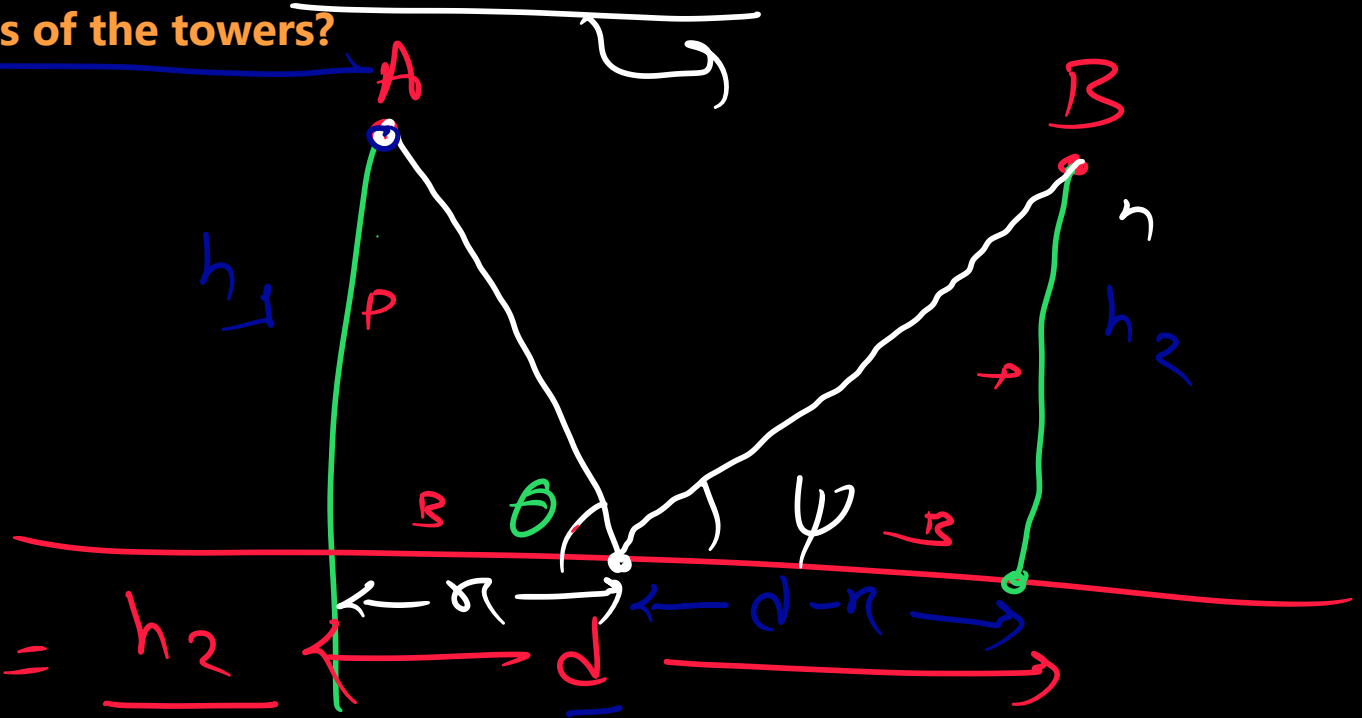
1. d (tan θ − tan φ)
2. x tan θ − (d − x) tan φ
3. tan θ − tan φ · d
4. (tan θ − tan φ) / d



Observer between two towers — elevation

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$h_1 - h_2$

$\tan\theta = \frac{h_1}{x}$, $\tan\psi = \frac{h_2}{-x}$

$x\tan\theta - (d-x)\tan\psi$ → (b)

Nearer tower

Observer is x metres from the nearer tower. Elevation to that top is θ.

$$\tan\theta = h_1 / x$$

Opposite = height. Adjacent = x.

$$h_1 = x \tan\theta$$

Next: the farther tower.

Farther tower, then the difference

Observer is (d − x) metres from the farther tower. Elevation to that top is φ.

$h_2 = (d - x) \tan \varphi$

Same idea. Adjacent is now d − x.

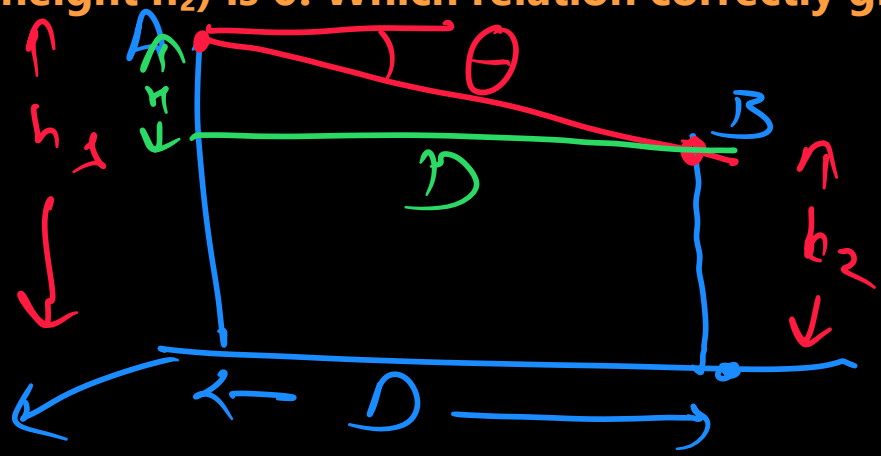
$h_1 - h_2 = x \tan \theta - (d - x) \tan \varphi$

If the question asks for the farther minus the nearer, flip the sign. Always sketch before you write tan.

From the top of one tower — depression

Two towers are separated by a horizontal distance D. From the top of the first tower (height h_1), the angle of depression to the top of the second tower (height h_2) is θ. Which relation correctly gives D in terms of h_1, h_2 and θ? Take $h_1 > h_2$.

1. $D = (h_1 + h_2) / \tan\theta$
2. $D = (h_1 - h_2) / \tan\theta$
3. $D = (h_2 - h_1) / \tan\theta$
4. $D = \tan\theta \cdot (h_1 + h_2)$



$x = h_1 - h_2$

Horizontal axis

$D = \dfrac{h_1 - h_2 = x}{\tan\theta}$

$\tan\theta = \dfrac{x}{D} \rightarrow D = \dfrac{x}{\tan\theta}$

$D = \dfrac{h_1 - h_2}{\tan\theta}$

Write tan θ first

From the top of the first tower, look down to the top of the second. $h_1 > h_2$.

$\tan\theta = (h_1 - h_2) / D$

Opposite = height drop. Adjacent = D.

Depression looks down, so the opposite side is the difference of heights, never the sum.

Next: solve for D.

Solve for D

$\tan\theta = (h_1 - h_2) / D$

$D = (h_1 - h_2) / \tan\theta$

Wrong options use $(h_1 + h_2)$ or multiply by $\tan\theta$. Depression looks down, so the opposite side is the difference of heights, never the sum.

Which ratios are positive?

A ratio is positive only in the quadrants named below. Reciprocals take the same sign.

I **0° to 90°** → 0 - 90 → All sin, cos, tan and the three reciprocals

II **90° to 180°** → Silver sin and cosec only

cos, tan, cot & cosec → −ve

III **180° to 270°** Tea tan and cot only

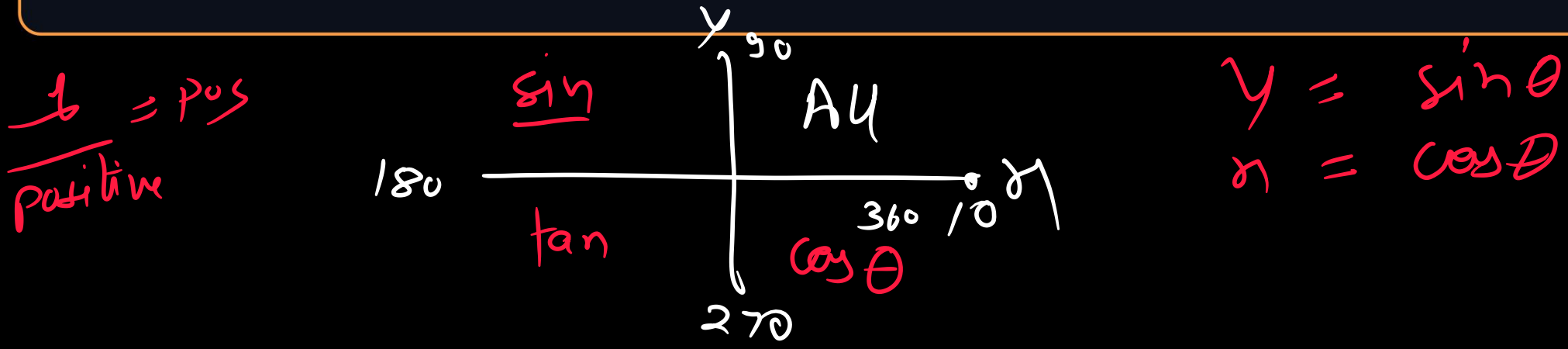
IV **270° to 360°** Cups cos and sec only

sin US =

Which ratios are positive?

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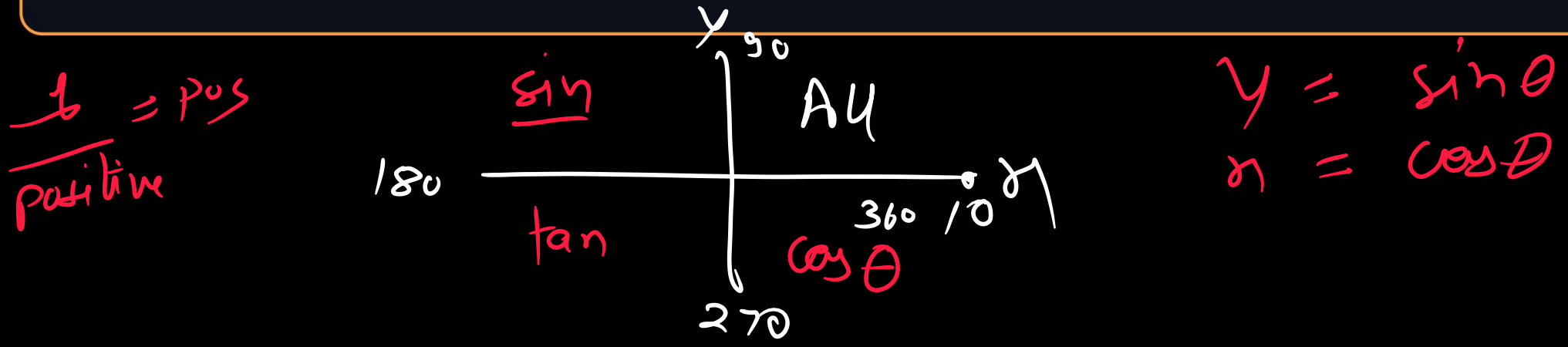
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All · Silver · Tea · Cups

cos

VIP

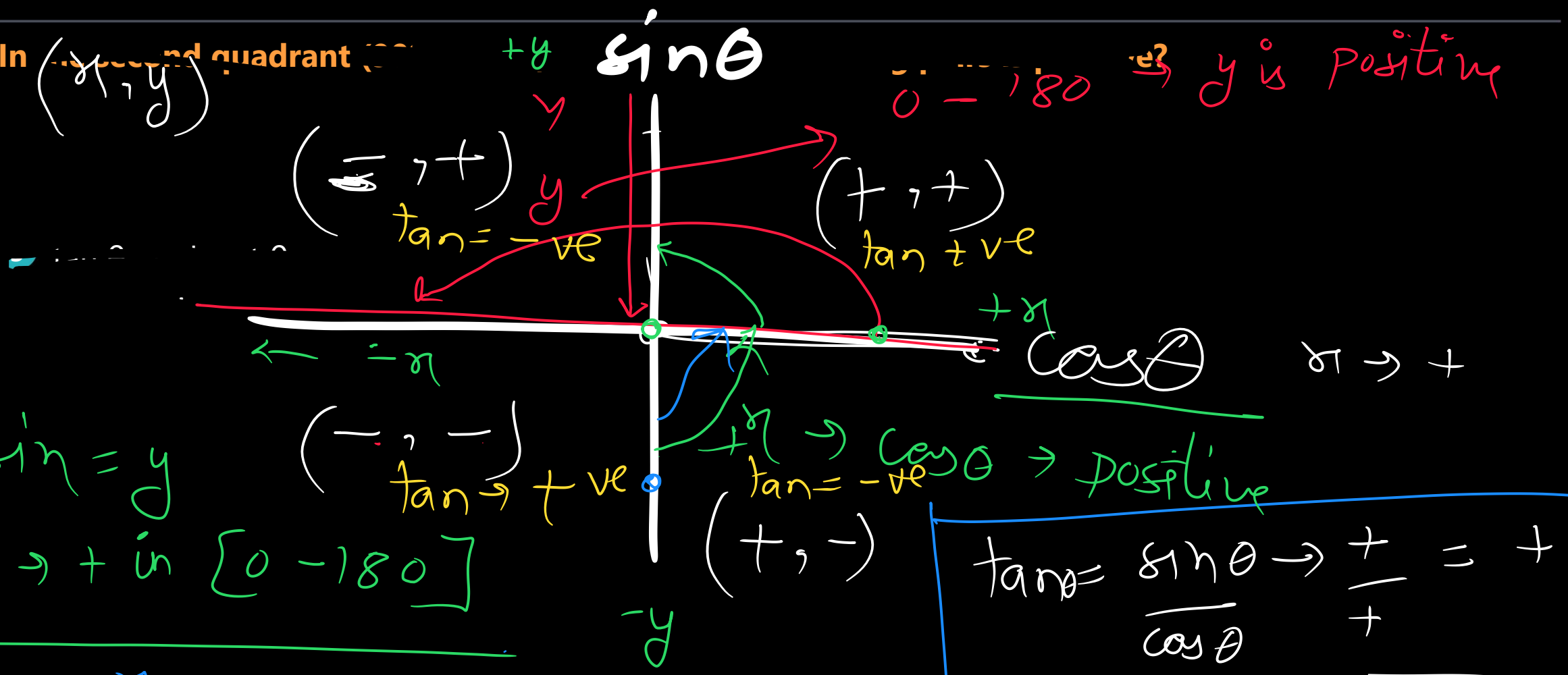
Write this as a plus-sign of four rooms. Start at the first quadrant and go anti-clockwise.

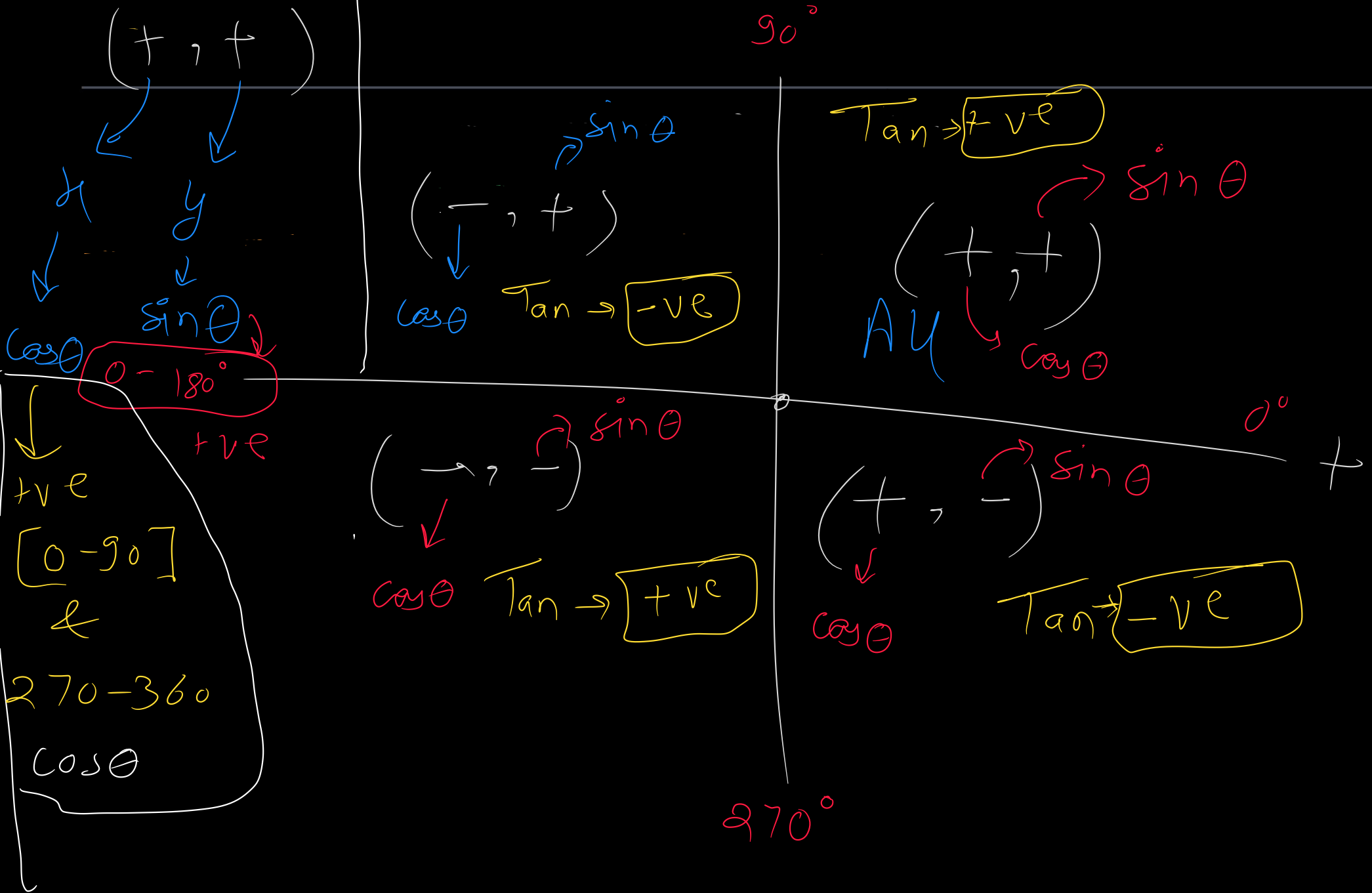
A S T C → **All sin tan cos**

Second quadrant: sine is positive, cosine is negative, so tangent is negative.
That is why cos 120° and tan 135° carry a minus in the table.

Signs in the second quadrant

In second quadrant




(+, +)
x
y
cosθ
sinθ
0 - 180°
+ve
+ve
[0-90]
&
270-360
cosθ
90°
(−, +)
sinθ
cosθ
Tan → -ve
Tan → +ve
(+, +)
sinθ
cosθ
All
(−, −)
sinθ
cosθ
Tan → +ve
0°
+
(+, −)
sinθ
cosθ
Tan → -ve
270°

Second quadrant — Silver

ASTC: second quadrant is S — sine. Cosecant is 1/sine, so it is positive too.

sin θ and cosec θ are positive. cos, sec, tan, cot are negative.

Do not say “sine is positive so tangent is positive”. Tangent is sine over cosine, and cosine is negative here.

sin, cos, tan from 0° to 90°

θ	0°	30°	45°	60°	90°
sin	0	1/2	1/√2	√3/2	1
cos	1	√3/2	1/√2	1/2	0
tan	0	1/√3	1	√3	undef.

tan 90° is not defined.

$\sin\theta = \cos(90-\theta) \rightarrow$

If $\sin 37° = 3/5$

$\theta = 37°$

$90 - \theta = 53$

If $\sin 37 = 3/5$

$\cos 53 = ?$

$\hookrightarrow \cos(90-37) = \sin 37 = 3/5$

sin, cos, tan from 0° to 90°

θ	0°	30°	45°	60°	90°
sin	0	1/2	1/√2	√3/2	1
cos	1	√3/2	1/√2	1/2	0
tan	0	1/√3	1	√3	undef.

tan 90° is not defined.

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \Rightarrow \tan 0 = \frac{\sin 0}{\cos 0} = \frac{0}{1} = 0$$

$$\tan 30 = \frac{\sin 30}{\cos 30} = \frac{1/2}{\sqrt{3}/2} = \frac{1}{\sqrt{3}}$$

$$\tan 60 = \frac{\sin 60}{\cos 60} = \frac{\sqrt{3}/2}{1/2} = \sqrt{3}/1 = \sqrt{3}$$

sin, cos, tan from 0° to 90°

θ	0°	30°	45°	60°	90°
sin	0	1/2	1/√2	√3/2	1
cos	1	√3/2	1/√2	1/2	0
tan	0	1/√3	1	√3	

tan 90° is not defined.

$\tan 60 = \frac{1}{\frac{\sqrt{3}}{2}} \Rightarrow \frac{2}{\sqrt{3}}$

$\sec 60 = \frac{1}{1/2} = 2$

$\tan 90 = \frac{1}{0} = \text{Undefined}$

$= \infty = \frac{1}{0}$

$\sec 90 = \frac{1}{0} = \infty$

sin, cos, tan from 90° to 180°

θ	90°	120°	135°	150°	180°
sin	1	√3/2	1/√2	1/2	0
cos	0	−1/2	−1/√2	−√3/2	−1
tan	undef.	−√3	−1	−1/√3	0

Range: −1 ≤ sin θ ≤ 1 and −1 ≤ cos θ ≤ 1

$$\sin\theta = \sin(180 - \theta)$$

Brother angle in the second quadrant

An angle past 90° uses a brother angle in the first quadrant. ASTC gives the sign.

sin (180° − θ) = sin θ
cos (180° − θ) = − cos θ
tan (180° − θ) = − tan θ

120° = 180° − 60°

Example: sin 120° = sin 60° = √3/2. cos 120° = − cos 60° = −1/2.

$\sin\theta = \cos(90 - \theta)$

$\sin\theta = \sin(180 - \theta)$

$\cos\theta = -\cos(180 - \theta)$

Read these from the table

From the 0° to 180° table, find the value of each of the following.

(i) sin 120°
(ii) cos 150°
(iii) tan 135°
(iv) sin 150°
(v) cos 120°

$\sin(120) = \sin(180 - \theta)$

$180 - \theta = 120$

$\theta = 60$

H.W $= \sin 60 = \sqrt{3}/2$

From the 0°–180° table

$\sin 120° = \sqrt{3} / 2$

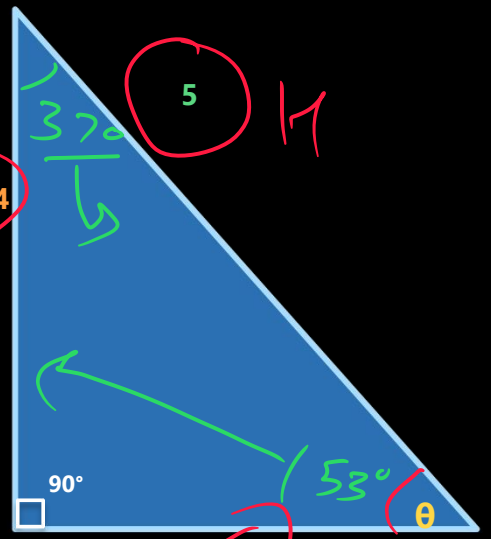
$\cos 150° = -\sqrt{3} / 2$

$\tan 135° = -1$

$\sin 150° = 1 / 2$

$\cos 120° = -1 / 2$

3–4–5 triangle — 37° and 53°



In a 3–4–5 right triangle, find sin, cos and tan of 37°, and sin, cos and tan of 53°.

$\Rightarrow 3^2 + 4^2 = ?$

$\downarrow$

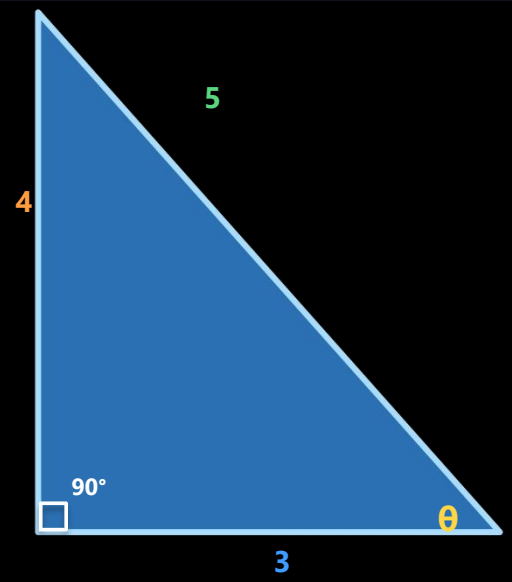
$9 + 16 = 25$

$= 5^2$

$\sin 37^\circ = \frac{3}{5}$

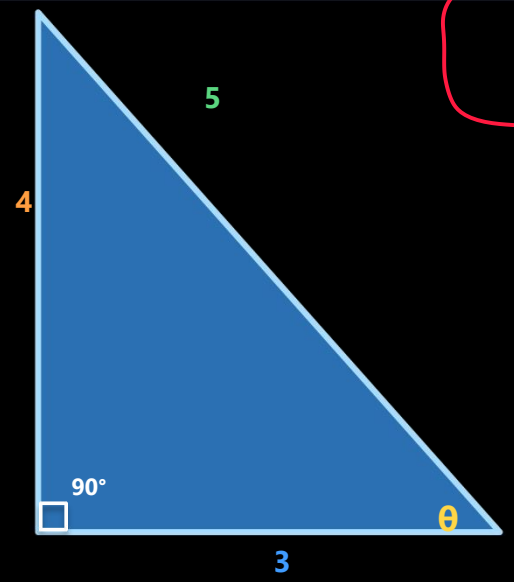
$\cos 37^\circ = \frac{B}{H} = \frac{4}{5}$

3–4–5 triangle — 37° and 53°



In a 3–4–5 right triangle, find sin, cos and tan of 37°, and sin, cos and tan of 53°.

3–4–5 triangle — 37° and 53°



sin 37° = 3/5 **cos 37° = 4/5** **tan 37° = 3/4**

sin 53° = 4/5 **cos 53° = 3/5** **tan 53° = 4/3**

tan 37° < tan 53° because 3/4 < 4/3.

Also: sin 53° × cosec 53° = 1.

Cos 53°

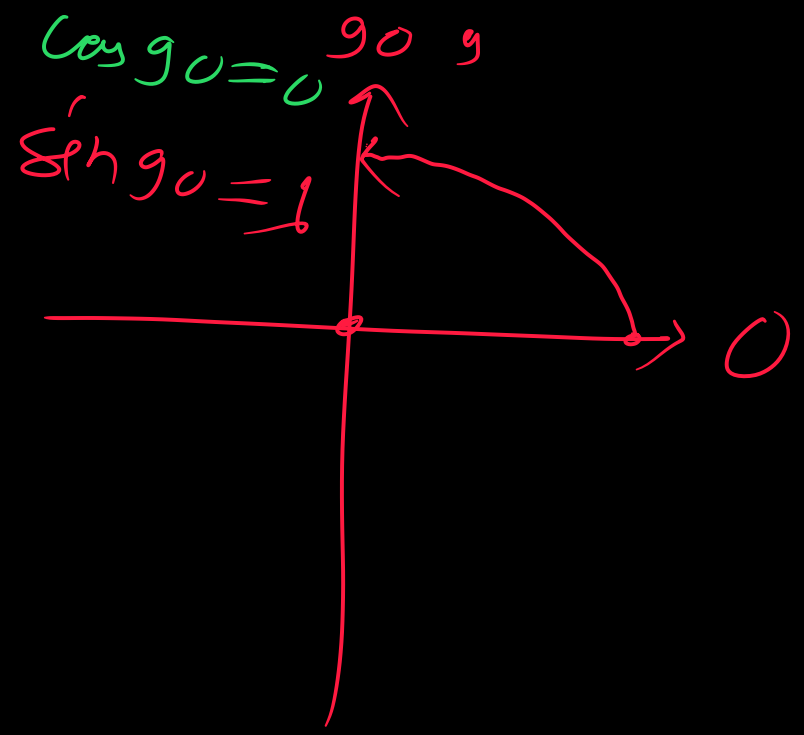
Tan 53°

Sin 53°

From 0° to 90°

If the angle increases from 0° to 90°, then what happens to the value of sin θ and the value of cos θ?

1. sin θ increases, cos θ decreases
2. sin θ decreases, cos θ increases
3. both decrease
4. both increase



sin 0 = 0

cos 0 = 1

From 0° to 90°

sin θ increases

$0 \rightarrow 1/2 \rightarrow 1/\sqrt{2} \rightarrow \sqrt{3}/2 \rightarrow 1$

cos θ decreases

$1 \rightarrow \sqrt{3}/2 \rightarrow 1/\sqrt{2} \rightarrow 1/2 \rightarrow 0$

0° → 90°: sin ↑ cos ↓ tan ↑ (tan 90° not defined)

Identity

Find the value of $\sin^2 40° + \cos^2 40°$.

$= ?$

$\rightarrow 1$

$\sin^2\theta + \cos^2\theta = 1$

$\sin^2 40° + \cos^2 40°$

True for every θ. Here θ = 40°.

$\sin^2\theta + \cos^2\theta = 1$

Do not compute 40°. Same angle, so the identity is enough.

$$\sin^2 40° + \cos^2 40° = 1$$

BREAK

Next: radian measure → conversions → tonight

BREAK

Next: radian measure → conversions → tonight

Arc length

Arc length = R θ

θ must be in radian. Never put 90° into this formula.

Practical unit of angle: degree. SI unit of angle: radian.

+θ anti-clockwise −θ clockwise

Arc = R θ

The one line: 180° = π rad

Lock this. Every conversion starts here.

180° = π rad

Next: degree → radian, then radian → degree.

Wherever you see π rad, put 180°.

Degree → radian

1° = π / 180 rad

Divide the lock-line by 180.

So θ° in radian = θ × (π / 180).

Multiply the degree by π/180.

Radian → degree

1 rad = 180° / π

Divide the lock-line by π.

So θ rad in degree = θ × (180 / π).

Multiply the radian by 180/π.

Arc on a circle

An object moves on the circumference of a circle of radius R. If the object turns through 90°, find the total distance moved by the object.

Object turns 90° on a circle of radius R

An object moves on the circumference of a circle of radius R and turns through 90°. Find the distance.

Wrong: Distance = R × 90°

Right: Distance = R × (π/2 rad) = πR/2

distance = path = arc = Rθ

Convert 90° first: 90° = π/2 rad

πR / 2

First conversion

Convert π/3 radian into degree.

π / 3 rad

Wherever you see π rad, put 180°.

π / 3 rad = 180° / 3 = 60°

π / 3 rad = 60°

Convert these now

Convert the following into degree or radian, as required.

(i) $\pi/2$ rad

(ii) $\pi/4$ rad

(iii) $\pi/6$ rad

(iv) 45°

(v) 90°

(vi) 60°

The six conversions from the board

$\pi/2$ rad = 90°

$\pi/6$ rad = 30°

90° = $\pi/2$ rad

$\pi/4$ rad = 45°

45° = $\pi/4$ rad

60° = $\pi/3$ rad

Find β

A triangle has two interior angles 40° and 70°. The side next to the 70° angle is extended. If β is the exterior angle formed at that vertex, find the value of β.

Convert radian into degree

Convert the following radian measures into degree.

(i) $5\pi/6$
(ii) $2\pi/3$
(iii) $4\pi/3$
(iv) $3\pi/5$
(v) 5π
(vi) $2\pi/5$
(vii) $7\pi/12$
(viii) $5\pi/2$
(ix) 3π

Convert degree into radian

Convert the following into radian.

(i) 120°
(ii) 30°
(iii) 150°
(iv) 270°
(v) 135°
(vi) π

Product of a ratio and its reciprocal

Find the value of $\sin 53° \times \operatorname{cosec} 53°$.

Product of three values

Find the value of $\sin 180° \cdot \cos 135° \cdot \tan 120°$.

Shadow of a pole

A vertical pole of height h casts a shadow of length l when the Sun is at an angle θ above the horizon. Which of the following expresses the shadow length l in terms of h and θ?

1. $l = h \tan \theta$
2. $l = h / \tan \theta$
3. $l = h \sin \theta$
4. $l = h / \sin \theta$

Wire from the top of a pole

A pole of height h has a wire attached from the top to the ground at an angle θ. If the wire breaks and falls to the ground without changing length, find the horizontal distance x from the base of the pole where the wire touches the ground.

1. $x = h \cot \theta$
2. $x = h \tan \theta$
3. $x = h \sin \theta$
4. $x = h \sec \theta$

Shadow on sloping ground

A vertical pole of height $h = 10$ m stands on ground that slopes upwards at a constant angle $\alpha = 30°$ with the horizontal. If the Sun's angle of elevation above the horizontal is $\theta = 60°$, what is the length of the shadow cast by the pole on the sloping ground?

1. 5 m
2. 10 m
3. $10\sqrt{3}$ m
4. $10 / (\sqrt{3} - 1)$ m

After class

Daily test before the next lecture.

What we lock from today

Height and distance. Sketch first. Tangent is opposite over adjacent.

Elevation: $h = x \tan \theta$

Two towers: $h_1 - h_2 = x \tan \theta - (d - x) \tan \varphi$

Depression: $D = (h_1 - h_2) / \tan \theta$

Depression looks down. Opposite is the difference of heights, never the sum.

ASTC and the table

A S T C → All sin tan cos

Reciprocals take the same sign. $\sin(180° - \theta) = \sin\theta$. $\cos(180° - \theta) = -\cos\theta$.

$\sin 37° = 3/5$ $\cos 37° = 4/5$ $\tan 37° = 3/4$

$\sin 53° = 4/5$ $\cos 53° = 3/5$ $\tan 53° = 4/3$

$\tan 90°$ is not defined. $-1 \le \sin\theta \le 1$ and $-1 \le \cos\theta \le 1$.

Radian — the one lock-line

180° = π rad

Wherever you see π rad, put 180°.

degree → radian: multiply by π/180
radian → degree: multiply by 180/π
Arc length = R θ with θ in radian. Never R × 90°.

90° on a circle of radius R → distance πR/2

After class

Daily test before the next lecture.
Tonight: remaining conversions and the height–distance PYQs.

Do not leave ASTC and 180° = π rad unwritten.